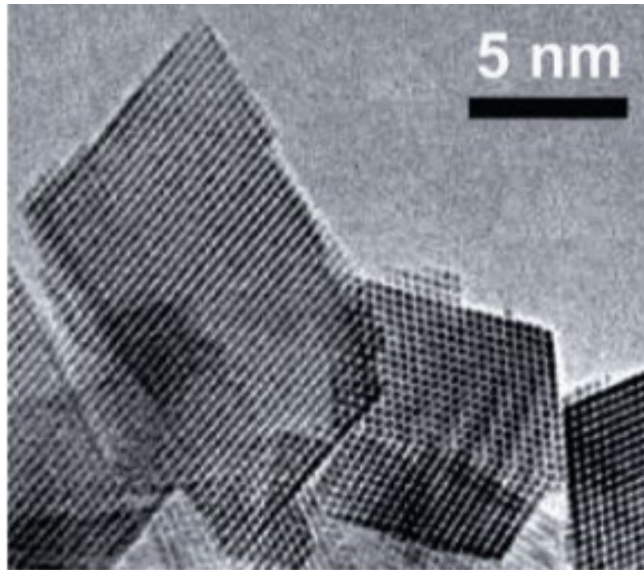


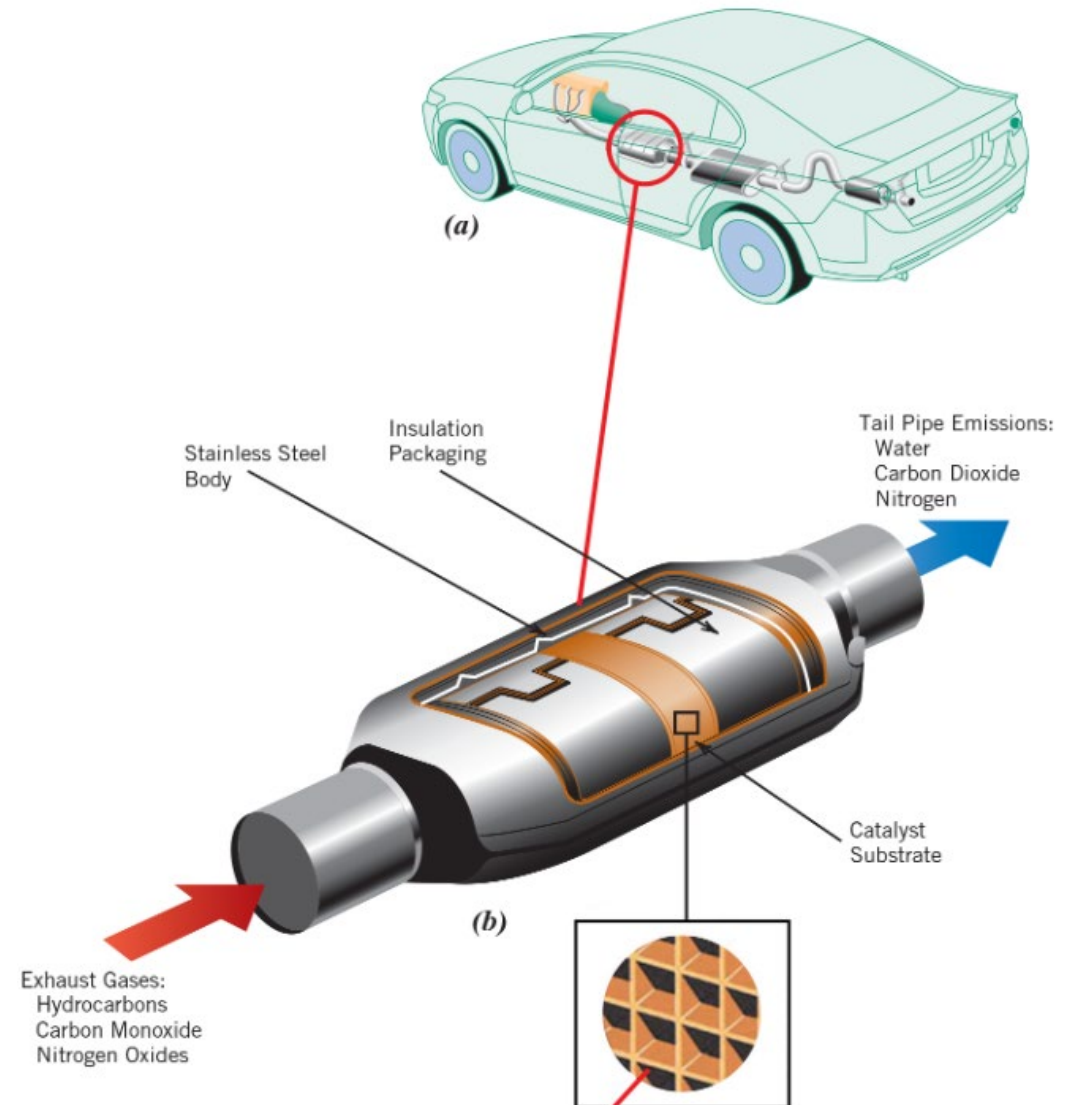


Imperfections in Solids

▣ Catalysts and Surface Defects



*High-resolution transmission electron micrograph showing surface defects on single crystals
($\text{Ce}_{0.5}\text{Zr}_{0.5}\text{O}_2$)*





Imperfections in Solids

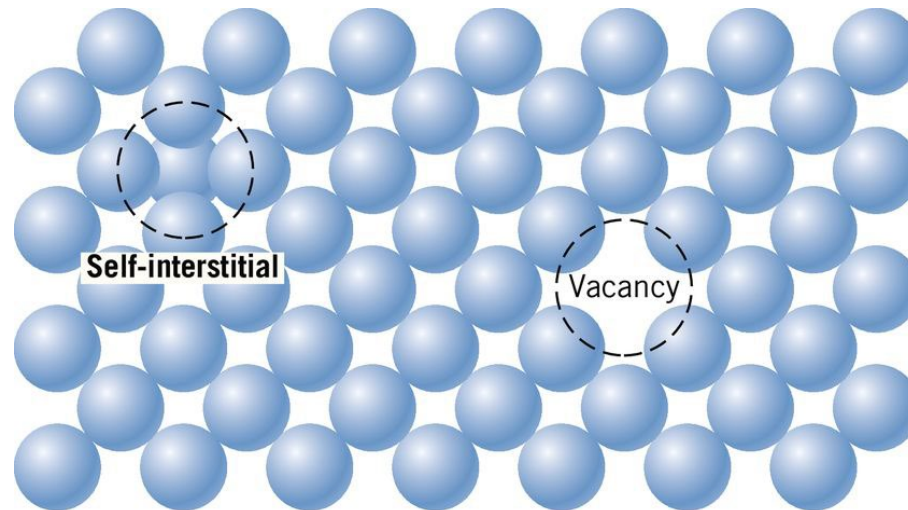
- ▣ Idealized crystalline solid with perfect atomic order *does not exist*; Crystalline **imperfections** (or **defects**) are always present
- ▣ Many properties of materials are sensitive to the *deviations* from crystalline perfection
- ▣ Crystalline imperfection can be classified by geometry or dimensionality of the defect
 - ▣ Point defects (0D)
 - ▣ Linear defects (1D)
 - ▣ Planar defects/interfacial defects/boundaries (2D)
 - ▣ Bulk/volume defects (3D)





Point Defects

- ▣ The simplest point defect is a **vacancy**, *i.e.*, an atom is missing
- ▣ **All** crystalline solids contain vacancies, as suggested by principles of thermodynamics (the existence of vacancies increases the entropy)
- ▣ A **self-interstitial** is an atom from the crystal that is crowded into an interstitial site, *i.e.* a small void space between original lattice





Point Defects

- ▣ Equilibrium number of vacancies N_v for a given quantity of material depends on and increases with temperature (T)

$$N_v = N e^{-\frac{Q_v}{kT}}$$

- ▣ Where N is the total number of atomic sites per volume (usually in m^3), Q_v is energy for vacancy formation, k is Boltzmann's constant, T is temperature
- ▣ N_v increases **exponentially** with T
- ▣ For most metals, at T just below T_{melting} , N_v/N is in the order of 10^{-4} , meaning 1 out of 10,000 lattice sites will be empty



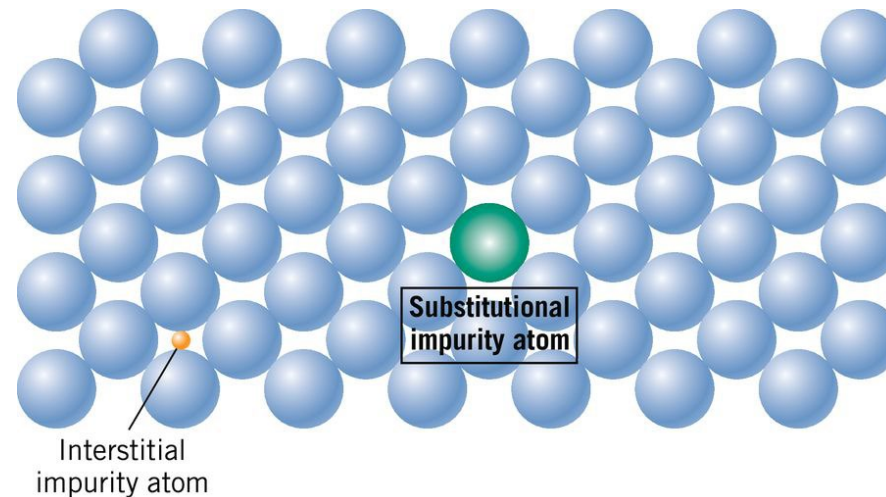
Impurities in Metals

- ▣ Impurity or foreign atoms are always present in metals
- ▣ Current refining techniques can hardly achieve purity $> 99.9999\%$
- ▣ Most familiar metal materials are **alloys**, in which impurity atoms were *intentionally* added to impart specific characteristics, such as mechanical strength, corrosion resistance, etc.
- ▣ Depending on the *kind* of impurity, the *concentration* of impurity, and the *temperature* of the alloy, addition of impurity atoms could result in formation of a **solid solution** and/or a new second phase



Solid Solution

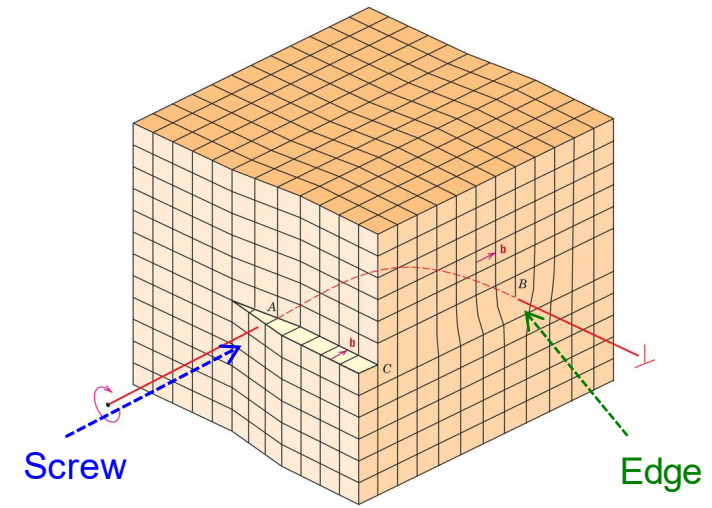
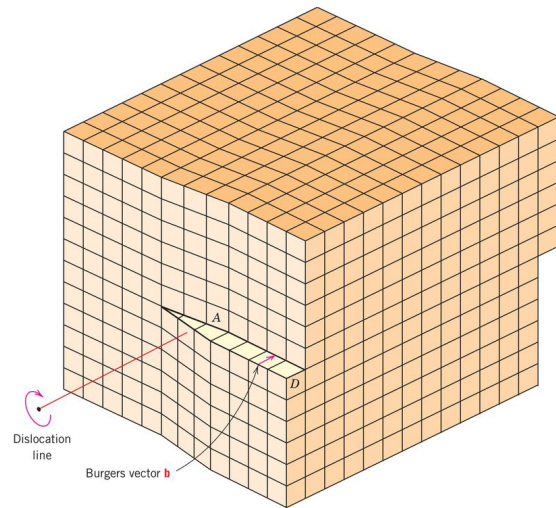
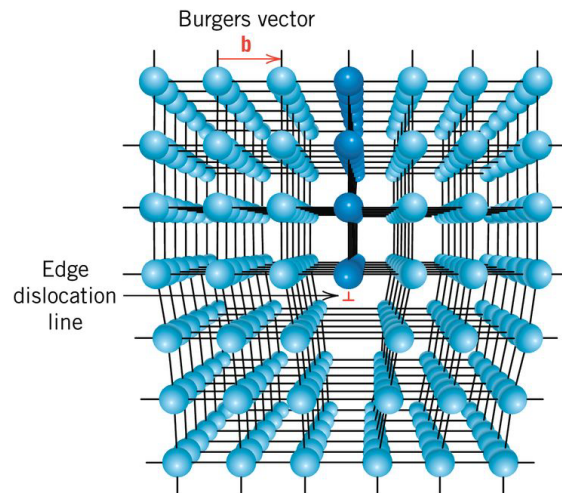
- ▣ A **solid solution** forms when the crystal structure is **maintained** and no new structures are formed; the impurity atoms are **randomly** and uniformly dispersed within the solid
- ▣ In **substitutional** solid solution, impurity atoms replace the host atoms
- ▣ In **interstitial** solid solution, impurity atoms fill the voids among the host atoms





Linear Defects

- ▣ A **dislocation** is a linear or one-dimensional defect around which some atoms are misaligned
- ▣ **Burgers vector** and **dislocation line** are often used to express dislocation

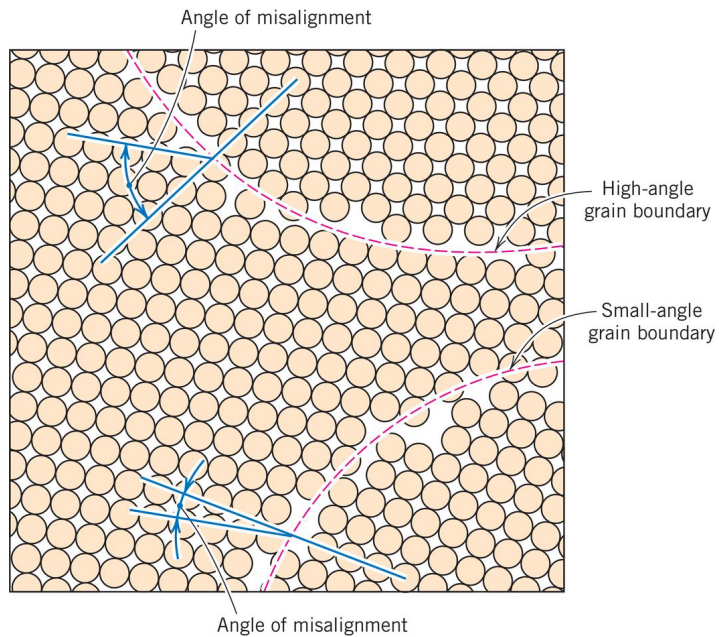


- ▣ **Edge dislocation**: formed by insertion of an extra portion of half-plane
- ▣ **Screw dislocation**: formed by a shear stress to distort the lattice
- ▣ **Mixed dislocation**

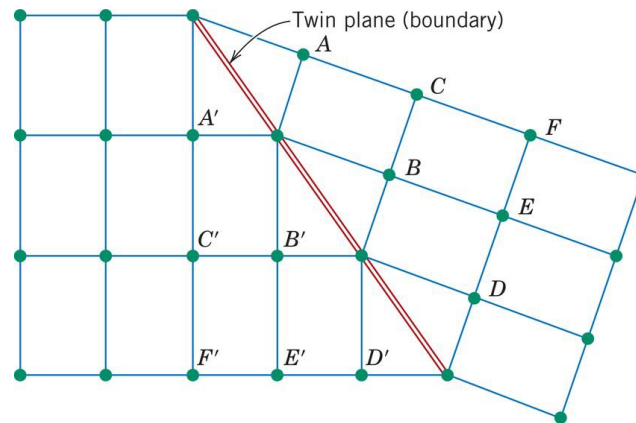


Interfacial Defects

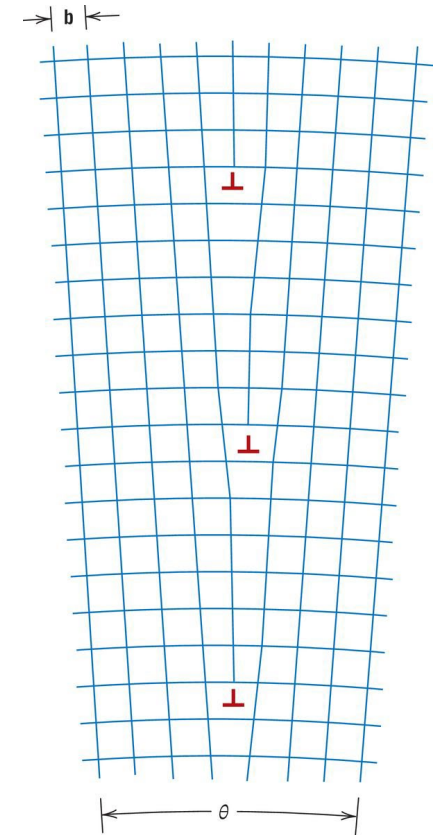
- ▣ **Interfacial defects** are **two-dimensional boundaries** that normally separate regions of materials having different crystal structures and/or crystallographic orientations



- ▣ **Grain boundary**



- ▣ **Twin boundary**: special grain boundary with mirror symmetry



- ▣ **Tilt boundary**: small-angle grain boundary where edge dislocations are aligned



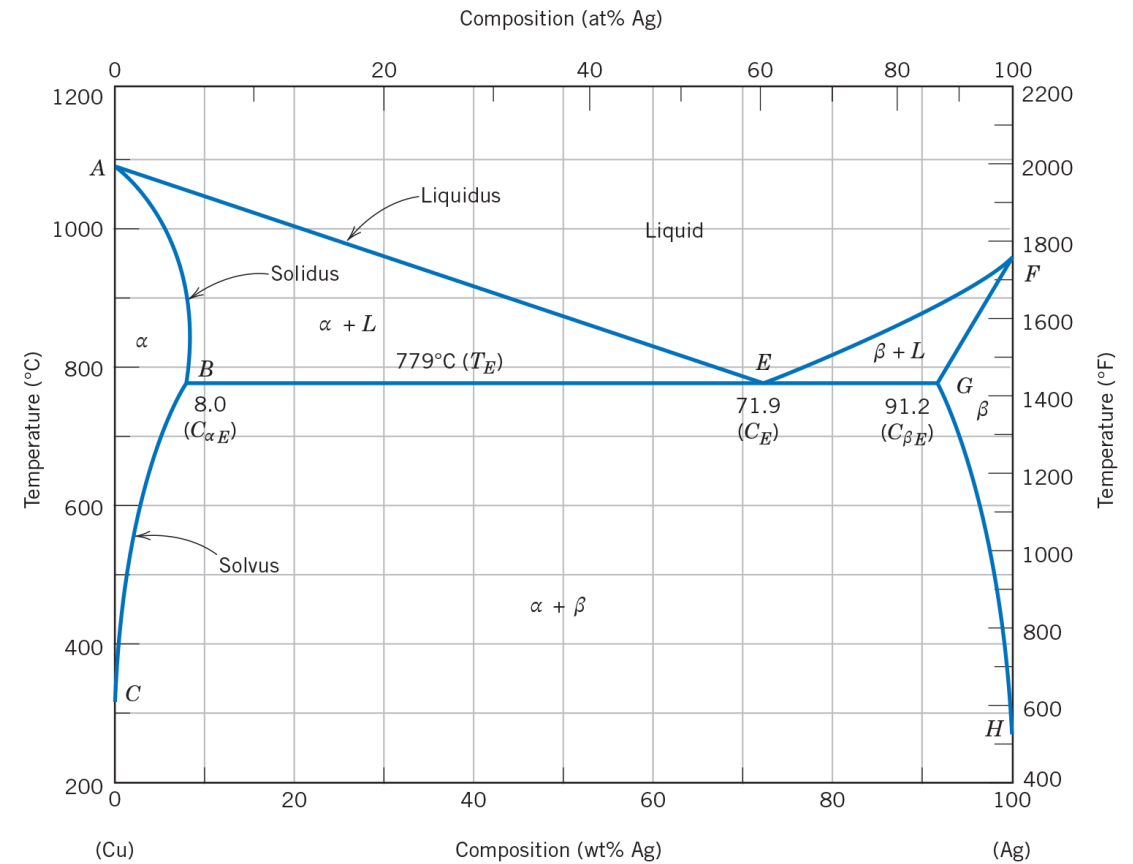
Interfacial Defects

▣ **External surface:** where crystal structure terminates.

- Bonds of surface atoms are not satisfied, giving rise to a surface energy

▣ **Phase boundary:** exist in multiphase materials, in which a different phase exists on each side of the boundary.

- Each phase has its own physical and/or chemical characteristics
- Every pure material is considered to be a phase; so also is every solid, liquid, and gaseous solution.

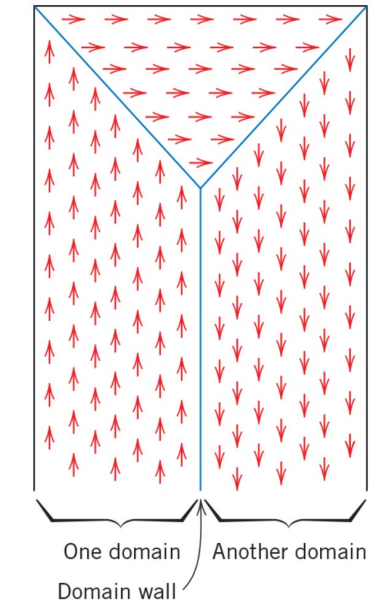
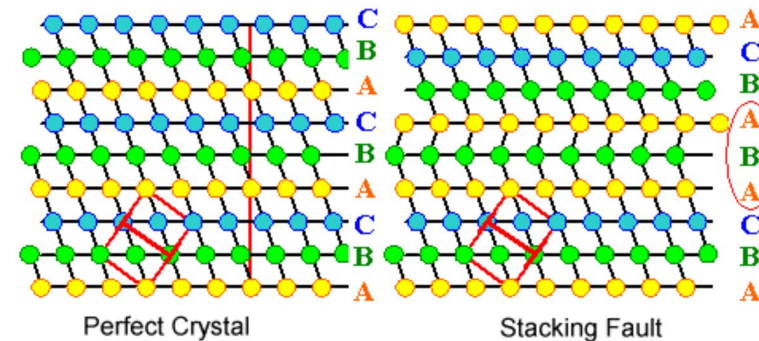


e.g. phase diagram of Cu-Ag alloy



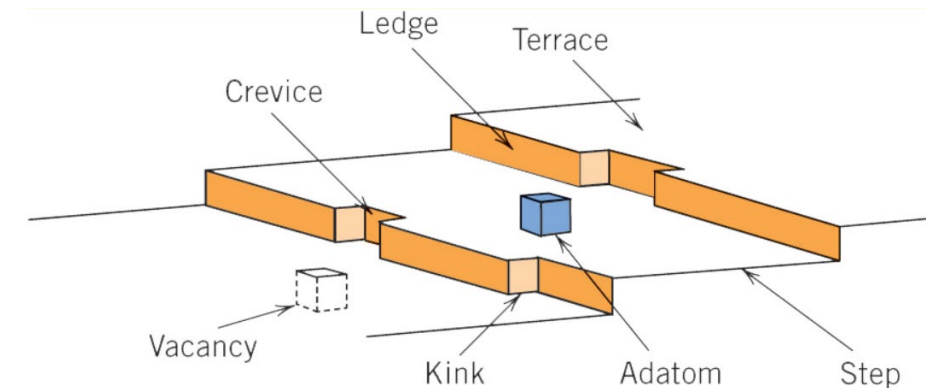
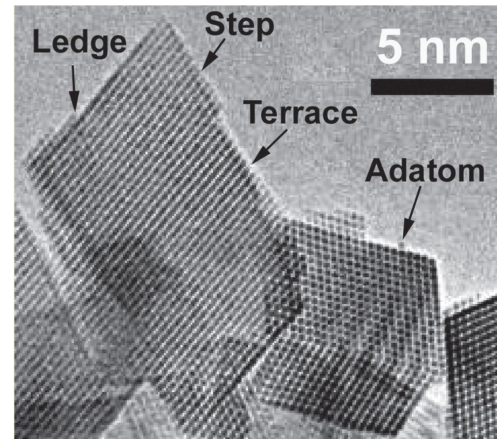
Interfacial Defects

- **Stacking faults**: exist often in FCC metals when “ABC” stacking sequence of close-packed planes is interrupted
- (ferromagnetic) **Domain wall**: boundary that separates regions with different directions of magnetization



Let's re-examine the catalyst surface we saw

Catalytic reactions normally occur at surface defect sites!





Bulk/Volume Defects

- ▣ There are also much larger defects in solid materials (usually in micrometer scales)
 - ▣ Pores
 - ▣ Cracks
 - ▣ Foreign inclusions
 - ▣ Other phases
- ▣ They are normally introduced during processing and fabrication

